PRACTICE TASKS

TASK 1 — CHECK YOUR IP ADDRESS

🎯 Objective

Find:

- IPv4 Address
- Default Gateway
- DNS Server

📌 Steps

Step 1

Press: Windows + R

Step 2 Type: cmd

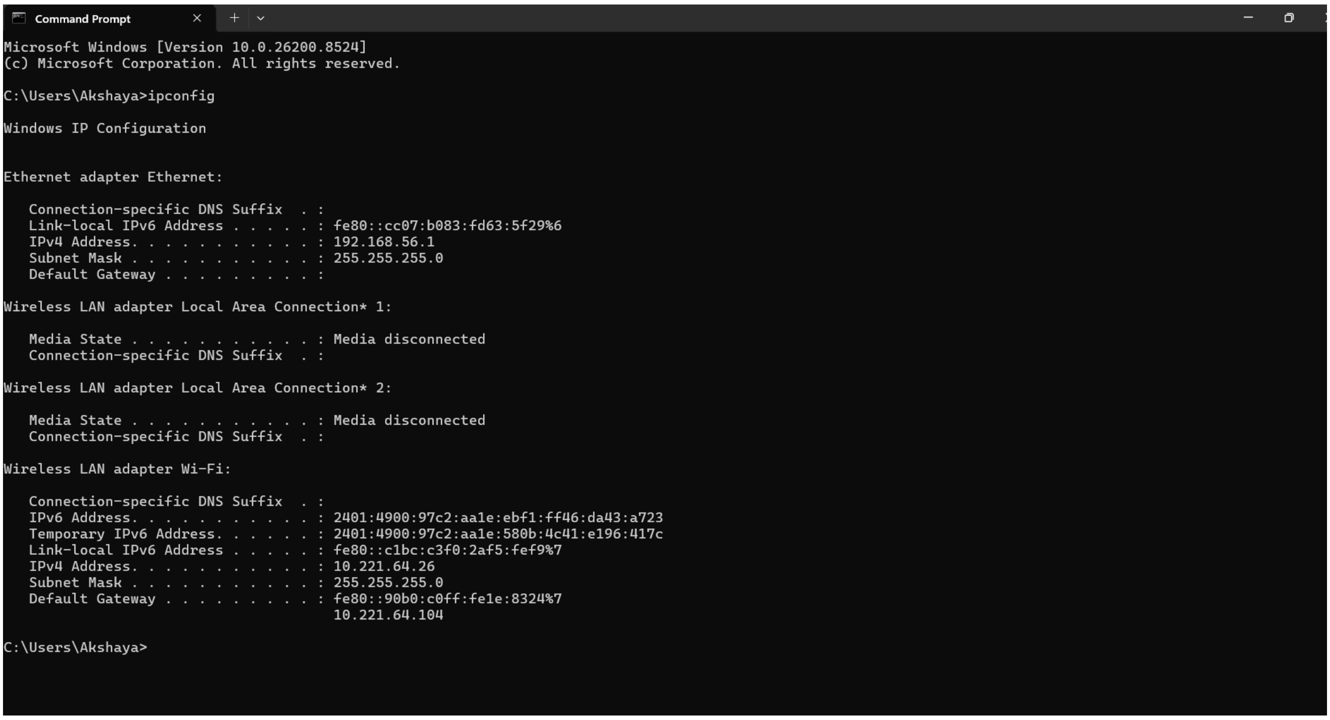
Press Enter.

Step 3 Command Prompt opens.

Type: ipconfig

Press Enter.

📸 Screenshot



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Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Akshaya>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::cc07:b083:fd63:5f29%6
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:97c2:aa1e:ebf1:ff46:da43:a723
   Temporary IPv6 Address. . . . . . : 2401:4900:97c2:aa1e:580b:4c41:e196:417c
   Link-local IPv6 Address . . . . . : fe80::c1bc:c3f0:2af5:fef9%7
   IPv4 Address. . . . . . . . . . . : 10.221.64.26
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::90b0:c0ff:fe1e:8324%7
                                       10.221.64.104

C:\Users\Akshaya>
```

NOTES

- ➢ **IPv4 Address** Identifies your device in network.

- ➢ **Default Gateway** Router address used to access internet.
- ➢ **DNS Server** Converts domain names into IP addresses.

TASK 2 — TEST INTERNET CONNECTIVITY

🎯 Objective

Check whether internet is working.

📌 Steps

In Command Prompt type:

ping google.com

Press Enter.

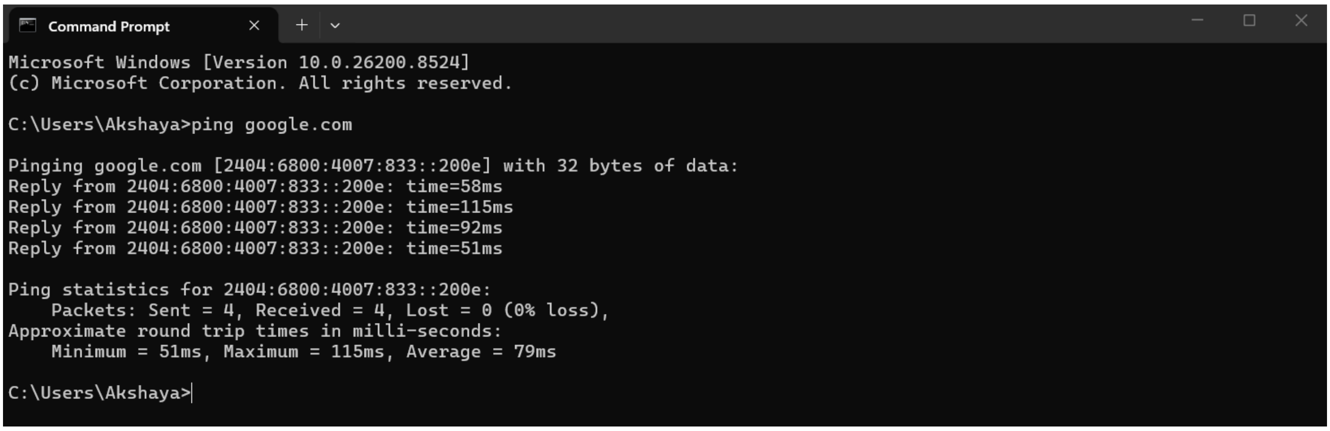
📌 Learn

Reply Packets: Shows connection success.

Time: Measures response speed.

Packet Loss: Shows network problems.

📸 Screenshot



Ping command checks:

- Connectivity
- Network availability
- Response time

TASK 3 — DNS LOOKUP

🎯 Objective

Understand how DNS converts names into IPs.

📌 Steps

Type:

nslookup google.com

Press Enter.

📌 Learn

DNS converts: google.com into : 142.x.x.x

📸 Screenshot

DNS helps users access websites using names instead of remembering IP addresses.

TASK 4 — IDENTIFY PRIVATE IP

🎯 Objective

Check whether your IP is private.

📌 Steps

Look at your IPv4 from:

ipconfig

📌 Compare With Private Ranges

192.168.x.x
10.x.x.x
172.16.x.x – 172.31.x.x

📌 Example

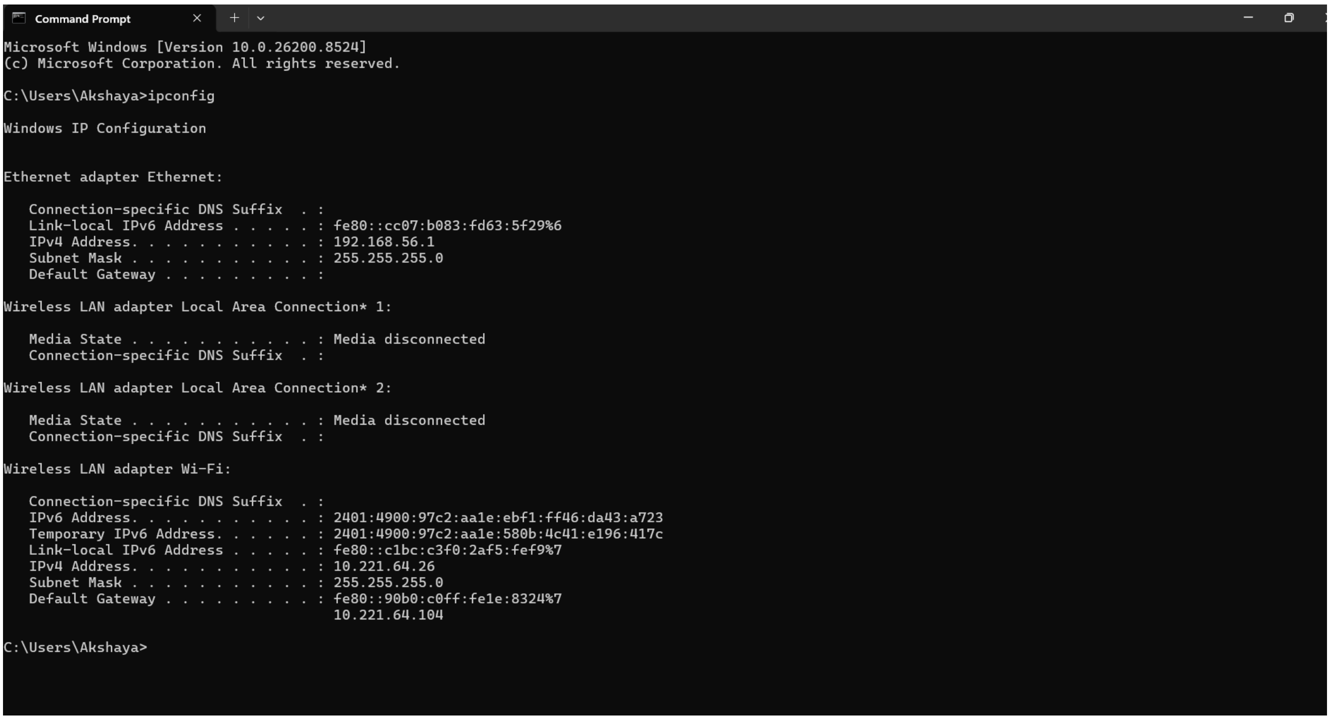
If your IP is:

192.168.1.5

Then it is a:

✅ Private IP

📸 Screenshot



Private IP addresses are used inside local networks and cannot directly access the internet.

TASK 5 — WRITE 5 VALID IPv4 ADDRESSES

🎯 Objective

Understand IPv4 structure.

📌 Steps

Write any 5 valid IPv4 addresses.

Example:

192.168.1.1
10.0.0.5
172.16.1.20
8.8.8.8
192.168.10.25

IMPORTANT NOTES

IPv4 contains:

- 4 octets
- Values from 0–255

TASK 6 — RESEARCH PUBLIC IP

🎯 Objective

Understand Public IP vs Private IP.

📌 Steps

Open browser.

Search:

What is my IP

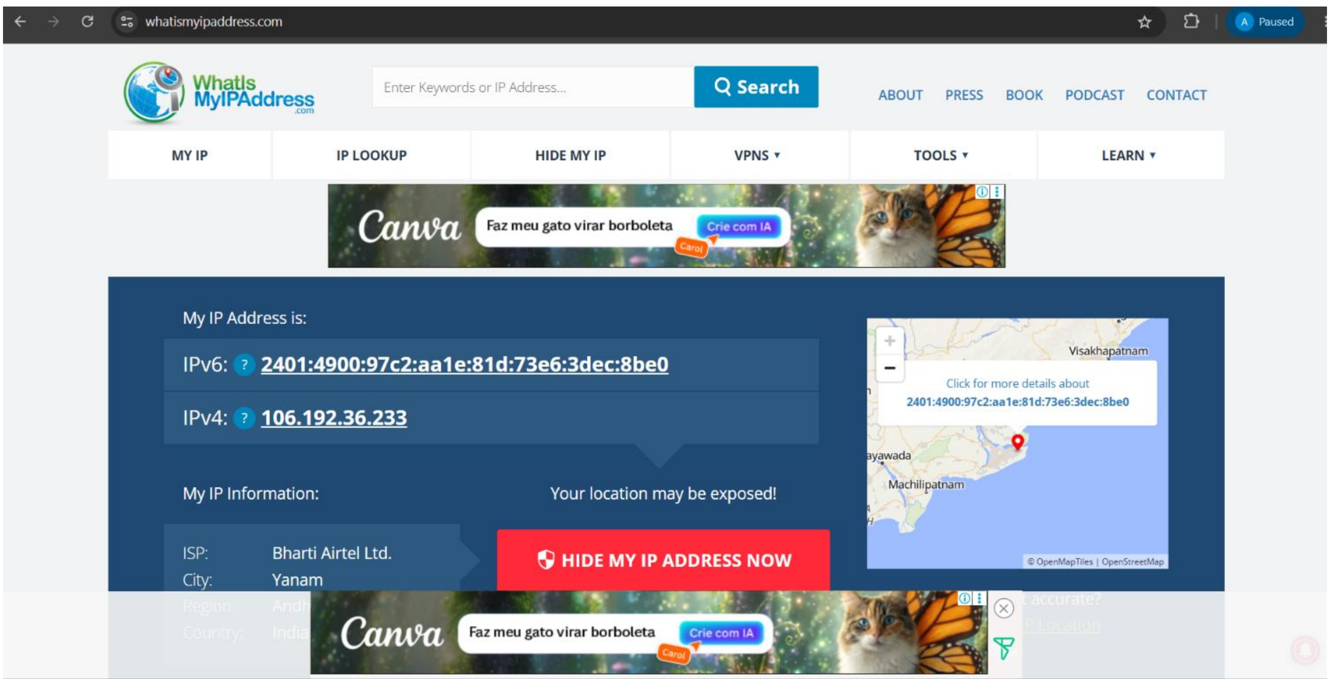
Google shows your public IP.

📌 Learn

Public IP: Used on internet.

Private IP: Used inside local network.

📸 Screenshot



Public IP is assigned by ISP and visible on internet.

TASK 7 — COMMON TERMS

🎯 Objective

Learn basic networking terminology.

📌 Write Meanings

IP: Unique address for devices.

DNS: Converts names into IP addresses.

Gateway: Connects network to internet.

Router: Routes traffic between networks.

Network: Group of connected devices.

TASK 8 — IAM CONNECTION TASK

🎯 Objective

Understand networking importance in IAM.

📌 Write:

"How IP Addresses Help IAM Security"

Minimum 5 points.

📝 Example Points

1. Track user login locations
2. Detect suspicious logins
3. Apply conditional access
4. Restrict unauthorized access
5. Monitor authentication activities